

Classification of the Orbits of Three Quadratic Collatz-type Maps

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Abstract

We study three quadratic Collatz-type maps on the non-negative integers. Each of them halves an even number, and they send an odd n to $n(n-1)/2$, to $n(n+1)/2$ and to $(n^2-1)/4$, respectively. Along an orbit, the odd terms grow until they reach a number of the form $2^m + 1$ or $2^m - 1$, depending on the map, and an orbit with an odd term greater than 1 is bounded exactly when this happens. The step that reaches such a number gives a solution of an exponential Diophantine equation, which becomes the equation $2^a \pm 2^b + 1 = z^2$ after completing the square. With Szalay's classification of the solutions of this equation, we classify all orbits of the maps. Every orbit either is eventually periodic or tends to infinity, and we determine exactly the initial values with bounded orbits. For the map with odd branch $n(n-1)/2$, Sedaghat stated the classification, and we give a new and complete proof. For this map, we also give an elementary criterion for unbounded orbits, based on divisibility alone. To our knowledge, the classifications for the other two maps are new.

1 Introduction

On the set $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ of non-negative integers, consider the maps

$$Q_-(n) = \begin{cases} \frac{n}{2}, & n \text{ even,} \\ \frac{n(n-1)}{2}, & n \text{ odd,} \end{cases} \quad Q_+(n) = \begin{cases} \frac{n}{2}, & n \text{ even,} \\ \frac{n(n+1)}{2}, & n \text{ odd,} \end{cases} \quad S(n) = \begin{cases} \frac{n}{2}, & n \text{ even,} \\ \frac{n^2-1}{4}, & n \text{ odd.} \end{cases} \quad (1)$$

The orbit $(x_i)_{i \geq 0}$ of an initial value $x_0 \in \mathbb{N}_0$ under Q_- is defined by $x_{i+1} = Q_-(x_i)$, so that $x_i = Q_-^i(x_0)$, where Q_-^i denotes the i -fold composition of Q_- . The orbits under Q_+ and S and the compositions Q_+^i and S^i are defined in the same way. For $x_0 \geq 1$, we write $x_0 = 2^l q$ with $l \geq 0$ and q odd, and call q the odd part of x_0 . In its compressed form, the Collatz map sends an even n to $n/2$ and an odd n to $(3n+1)/2$. As in the study of the Collatz map, we ask which initial values have bounded orbits, where these orbits end, and whether all other orbits tend to infinity.

Sedaghat [2] studies Q_- as a quadratic analogue of the Collatz-type map that sends an even n to $n/2$ and an odd n to $(3n-1)/2 = n + (n-1)/2$, and calls Q_- the divide-or-choose-2 rule. In Q_- and Q_+ , the sums $n + (n-1)/2$ and $(3n+1)/2 = n + (n+1)/2$ are replaced by the products $n \cdot (n-1)/2$ and $n \cdot (n+1)/2$. Sedaghat [2, Section 1] states that the orbits of Q_+ are qualitatively similar to those of Q_- , and does not study Q_+ further. The odd branch of S is the product of the consecutive integers $(n-1)/2$ and $(n+1)/2$, and Sedaghat [2, Section 1] calls S the symmetric rule. For Q_- , Sedaghat conjectures that unbounded orbits exist, and that every orbit containing an odd number $2^m - 1$ with $m \geq 3$ is unbounded [2, Conjectures 3 and 4]. For S , the existence of unbounded orbits is left open in [2, Section 4].

Sedaghat [3, Theorem 3] states that the orbit of $2^l(2^m + 1)$ with $l, m \geq 1$ under Q_- ends in a cycle of length m , and that every other orbit goes to infinity. The proof there reduces the classification to the statement that $k(2^j k + 1) = 2^m + 1$ has no solution in integers $j, m \geq 1$ and odd $k \geq 3$ [3, Lemma 2]. This statement is proved there by a descent on powers of 2. The proof does not treat the case in which the descent stops (Remark 1). We are not aware of other work on Q_+ or on the classification of the orbits of S .

map	odd branch	terminal numbers	equation (3)	exceptional odd parts	theorem
Q_-	$n(n-1)/2$	$2^m + 1$	$k(2^j k + 1) = 2^m + 1$	none	Theorem 1
Q_+	$n(n+1)/2$	$2^m - 1$	$k(2^j k - 1) = 2^m - 1$	5	Theorem 2
S	$(n^2 - 1)/4$	$2^m \pm 1$	$k(2^j k + \varepsilon) = 2^m + \delta$	11, 23, 181	Theorem 3

Table 1: The Collatz-type maps of (1): the odd branch, the terminal numbers with $m \geq 1$, the form of (3) that arises in Lemma 3, the odd parts $q > 1$ that are not terminal numbers but have a bounded orbit, and the classification theorem.

Our approach starts from the terminal numbers defined below, which differ between Q_- , Q_+ and S and decide whether an orbit with an odd term greater than 1 enters a cycle. Let $y > 1$ be an odd term of an orbit. We write

$$y = \begin{cases} 2^j k + 1, & \text{under } Q_-, \\ 2^j k - 1, & \text{under } Q_+, \\ 2^{j+1} k + \varepsilon, & \text{under } S, \end{cases} \quad j \geq 1, \quad k \text{ odd}, \quad (2)$$

where $\varepsilon \in \{-1, 1\}$ is chosen under S such that 4 divides $y - \varepsilon$. We also set $\varepsilon = 1$ under Q_- and $\varepsilon = -1$ under Q_+ . For each odd $y > 1$, the numbers j , k and ε in the decomposition (2) are unique.

In each case, the next odd term after y in the orbit is $y' = k(2^j k + \varepsilon)$ (Lemma 1). If $k \geq 3$, then $y' > y$. If $k = 1$, then y is a number $2^m + 1$ under Q_- , a number $2^m - 1$ under Q_+ , and a number $2^m \pm 1$ under S . We call these numbers, with $m \geq 1$, the *terminal numbers* of the map. They lie on cycles under Q_- and Q_+ , and their orbits reach 0 under S . Hence, if the odd part of x_0 exceeds 1, the orbit is bounded if and only if its odd terms reach a terminal number (Lemma 2). If the odd part of x_0 is not a terminal number but a later odd term is, the earliest step that reaches a terminal number $2^m + \delta$ gives a solution of

$$\boxed{k(2^j k + \varepsilon) = 2^m + \delta, \quad j, m \geq 1, \quad k \geq 3 \text{ odd}, \quad \varepsilon, \delta \in \{-1, 1\}}, \quad (3)$$

with $\varepsilon = \delta = 1$ under Q_- , with $\varepsilon = \delta = -1$ under Q_+ , and with $\varepsilon, \delta \in \{-1, 1\}$ under S (Lemma 3). Multiplying (3) by 2^{j+2} and adding 1 gives

$$(2^{j+1} k + \varepsilon)^2 = 2^{m+j+2} + \delta 2^{j+2} + 1, \quad (4)$$

so (3) asks when a number $2^a \pm 2^b + 1$ is a square. This reduces the study of terminal numbers for the Collatz-type maps to a Diophantine equation.

Szalay [5] determined all squares of the form $2^a \pm 2^b + 1$ with $a > b \geq 1$. By this result, the solutions of (3) are

$$3 \cdot (2 \cdot 3 - 1) = 2^4 - 1, \quad 3 \cdot (4 \cdot 3 - 1) = 2^5 + 1, \quad 45 \cdot (2 \cdot 45 + 1) = 2^{12} - 1 \quad (5)$$

(Lemma 5). None of them has $\varepsilon = \delta = 1$, and only $3 \cdot (2 \cdot 3 - 1) = 2^4 - 1$ has $\varepsilon = \delta = -1$. We call an odd part $q > 1$ *exceptional* if it is not a terminal number but its orbit is bounded. By the decomposition (2), the solutions (5) give the exceptional odd part $5 = 2 \cdot 3 - 1$ of Q_+ and the exceptional odd parts $11 = 4 \cdot 3 - 1$, $23 = 8 \cdot 3 - 1$ and $181 = 4 \cdot 45 + 1$ of S (Table 1).

Our main results are the following theorems. They characterize all orbits of the Collatz-type maps Q_- , Q_+ and S in (1).

Theorem 1 Let $x_0 \in \mathbb{N}_0$, and let $(x_i)_{i \geq 0}$ be its orbit under Q_- .

- (i) If $x_0 = 0$ or $x_0 = 2^l$ with $l \geq 0$, the orbit reaches the fixed point 0.
- (ii) If $x_0 = 2^l(2^m + 1)$ with $l \geq 0$ and $m \geq 1$, the orbit enters the cycle of length m that contains $2^m + 1$.
- (iii) For every other x_0 , the orbit satisfies $x_i \rightarrow \infty$ as $i \rightarrow \infty$.

Theorem 2 Let $x_0 \in \mathbb{N}_0$, and let $(x_i)_{i \geq 0}$ be its orbit under Q_+ .

- (i) If $x_0 = 0$, the orbit stays at the fixed point 0.

- (ii) If $x_0 = 2^l(2^m - 1)$ with $l \geq 0$ and $m \geq 1$, the orbit enters the cycle of length m that contains $2^m - 1$, which for $m = 1$ is the fixed point 1.
- (iii) If $x_0 = 5 \cdot 2^l$ with $l \geq 0$, the orbit enters the cycle $15 \rightarrow 120 \rightarrow 60 \rightarrow 30 \rightarrow 15$ through $15 = 2^4 - 1$.
- (iv) For every other x_0 , the orbit satisfies $x_i \rightarrow \infty$ as $i \rightarrow \infty$.

Theorem 3 Let $x_0 \in \mathbb{N}_0$, and let $(x_i)_{i \geq 0}$ be its orbit under S .

- (i) If $x_0 = 0$, or if the odd part of x_0 is of the form $2^m \pm 1$ with $m \geq 1$ or lies in $\{11, 23, 181\}$, the orbit reaches the fixed point 0.
- (ii) For every other x_0 , the orbit satisfies $x_i \rightarrow \infty$ as $i \rightarrow \infty$.

Theorem 1 is the classification of Sedaghat [3, Theorem 3], with $l \geq 0$ and with the initial values 0 and 2^l . We provide a new and complete proof: we transform $k(2^j k + 1) = 2^m + 1$ into the equation $2^a + 2^b + 1 = z^2$, whose solutions were determined by Szalay [5], and conclude that $k(2^j k + 1) = 2^m + 1$ has no solution with odd $k \geq 3$ (Lemma 5 (i)). To our knowledge, Theorems 2 and 3 are new. Theorem 3 shows that S has unbounded orbits, for instance the orbit of 13. This answers the question left open by Sedaghat [2]. For each map, only $O((\log X)^2)$ initial values in $\{0, 1, \dots, X\}$ have a bounded orbit (Corollary 1).

According to Scott [1, Section 1], neither Szalay's proof for $2^a + 2^b + 1 = z^2$ nor the shorter proof of Scott is strictly elementary, so our proof of Theorem 1 is not strictly elementary. For Q_- , Theorem 4 gives an elementary criterion: if an odd term $y > 1$ of an orbit divides none of the numbers $2^m + 1$, then the orbit tends to infinity. Moreover, y divides such a number if and only if the multiplicative orders of 2 modulo the prime factors of y are even and have the same 2-adic valuation. The criterion gives an elementary proof of both conjectures of Sedaghat [2].

Section 2 studies the odd terms of an orbit under Q_- , Q_+ and S , and reduces the classification to (3) (Lemma 3). Section 3 solves (3) and discusses the descent of [3]. Section 4 proves Theorems 1, 2 and 3 and Corollary 1. Section 5 proves the divisibility criterion for Q_- and uses it to show, without Szalay's result, that the initial values with a bounded orbit under Q_- have natural density 0. It also explains why the criterion does not carry over to Q_+ and S , and why Theorem 2 has an elementary proof. Section 6 states open questions.

2 Odd terms of an orbit

Throughout this section, an odd number $y \geq 3$ is written as in (2), with the sign ε defined there.

Lemma 1 Let $y \geq 3$ be odd, and write y as in the decomposition (2).

- (i) $Q_-(y) = 2^{j-1}ky$ and $Q_-^j(y) = k(2^j k + 1)$.
- (ii) $Q_+(y) = 2^{j-1}ky$ and $Q_+^j(y) = k(2^j k - 1)$.
- (iii) $S(y) = 2^j k(2^j k + \varepsilon)$ and $S^{j+1}(y) = k(2^j k + \varepsilon)$.

That is, in each case the next odd term after y in the orbit is $y' = k(2^j k + \varepsilon)$, and the terms between y and y' are multiples of y' . If $k \geq 3$, then $y' > y$.

Proof. By (1) and the decomposition (2), $Q_-(y) = y \cdot (y - 1)/2$ with $y - 1 = 2^j k$, and $Q_+(y) = y \cdot (y + 1)/2$ with $y + 1 = 2^j k$. In both cases, this is $2^{j-1}ky$. The $j - 1$ halvings that follow take $2^{j-1}ky$ to the odd number $y' = ky = k(2^j k + \varepsilon)$ through multiples of y' . If $k \geq 3$, then $y' = ky > y$.

Under S , we have $y + \varepsilon = 2^{j+1}k + 2\varepsilon = 2(2^j k + \varepsilon)$, so

$$y^2 - 1 = (y - \varepsilon)(y + \varepsilon) = 2^{j+2}k(2^j k + \varepsilon),$$

which gives $S(y)$. The number $2^j k + \varepsilon$ is odd, since $j \geq 1$, so j halvings take $S(y)$ to the odd number $y' = k(2^j k + \varepsilon)$ through multiples of y' . If $k \geq 3$, then

$$y' - y = 2^j k(k - 2) + \varepsilon(k - 1) \geq 2k(k - 2) - (k - 1) > 0.$$

□

With k as in the decomposition (2), an odd number $y \geq 3$ is a terminal number of the map (Table 1) if and only if $k = 1$. We next describe the orbit of a terminal number.

Under Q_- and Q_+ , let $y = 2^m + \varepsilon \geq 3$ be a terminal number. Lemma 1 with $j = m$ and $k = 1$ gives $Q_-^m(y) = y$ under Q_- and $Q_+^m(y) = y$ under Q_+ . The orbit of y runs through the m distinct numbers $2^i(2^m + \varepsilon)$ with $0 \leq i \leq m - 1$, which form the cycle of length m

$$2^m + \varepsilon \rightarrow 2^{m-1}(2^m + \varepsilon) \rightarrow \cdots \rightarrow 2(2^m + \varepsilon) \rightarrow 2^m + \varepsilon. \quad (6)$$

For $m = 1$, this cycle is the fixed point 3 of Q_- . The terminal number 1 of Q_+ is also a fixed point, since $Q_+(1) = 1$.

Under S , let $y = 2^{j+1} + \varepsilon$ be a terminal number. Lemma 1 with $k = 1$ gives $y' = 2^j + \varepsilon$, which is again a terminal number. By repeated use of this step, the odd terms finally reach 1 or 3. Since S maps $3 \rightarrow 2 \rightarrow 1 \rightarrow 0$, the orbit of every terminal number reaches the fixed point 0. This is [2, Lemma 6].

The next lemma combines Lemma 1 with the orbits of the terminal numbers, and describes every orbit whose initial value has an odd part greater than 1.

Lemma 2 *Let $x_0 \geq 1$ have odd part $q > 1$, and let $(x_i)_{i \geq 0}$ be its orbit under Q_- , Q_+ or S .*

- (i) *Under Q_- and Q_+ , q divides x_i for every $i \geq 0$.*
- (ii) *Let $y_0 = q, y_1, y_2, \dots$ be the odd terms of the orbit, in order of occurrence. For $y_r \geq 3$, let j_r, k_r and ε_r be the numbers of the decomposition (2) for $y = y_r$. By Lemma 1,*

$$y_{r+1} = k_r(2^{j_r} k_r + \varepsilon_r). \quad (7)$$

If some y_r is a terminal number, then under Q_- and Q_+ the orbit enters the cycle (6) that contains y_r , and under S it reaches the fixed point 0. Otherwise $k_r \geq 3$ for every r , and $x_i \rightarrow \infty$ as $i \rightarrow \infty$.

Proof. Under Q_- and Q_+ , let q divide n . If n is even, q divides $n/2$, since q is odd. If n is odd, $Q_-(n) = n \cdot (n-1)/2$ and $Q_+(n) = n \cdot (n+1)/2$ are multiples of n . Hence q divides x_{i+1} whenever it divides x_i . Since q divides x_0 , induction on i proves (i).

The l halvings take x_0 to $q = y_0$. If some y_r is a terminal number, the claim follows from the discussion before this lemma, since (i) gives $y_r \geq q \geq 3$ under Q_+ . Assume now that no y_r is a terminal number. If $y_r \geq 3$, then $k_r \geq 3$, and Lemma 1 gives $y_{r+1} > y_r$. Since $y_0 = q \geq 3$, induction on r gives $k_r \geq 3$ and $y_{r+1} > y_r$ for every r . Hence $y_r \geq q + r$. By Lemma 1, every term after y_r is at least y_{r+1} , so $x_i \rightarrow \infty$. This proves (ii). □

For Q_- , Lemma 2 (ii) combines [2, Theorem 2] with an observation in the proof of [3, Theorem 3]: an orbit that does not enter a cycle tends to infinity. For S , it contains [2, Theorem 7].

By Lemma 2 (ii), it remains to determine the initial values whose orbits have an odd term that is a terminal number.

Lemma 3 *Let $x_0 \geq 1$ have odd part $q > 1$, and consider its orbit under Q_- , Q_+ or S . Assume that q is not a terminal number of the map, and that some odd term of the orbit is a terminal number. Let r be the least index such that y_r is a terminal number, where y_r is as in Lemma 2 (ii). Then $r \geq 1$, $k_{r-1} \geq 3$, and $j = j_{r-1}$, $k = k_{r-1}$ give the following solutions of (3).*

- (i) *Under Q_- , $y_r = 2^m + 1$ with $m \geq 1$, and $k_{r-1}(2^{j_{r-1}} k_{r-1} + 1) = 2^m + 1$.*
- (ii) *Under Q_+ , $y_r = 2^m - 1$ with $m \geq 1$, and $k_{r-1}(2^{j_{r-1}} k_{r-1} - 1) = 2^m - 1$.*
- (iii) *Under S , $y_r = 2^m + \delta$ with $m \geq 1$ and $\delta \in \{-1, 1\}$, and $k_{r-1}(2^{j_{r-1}} k_{r-1} + \varepsilon_{r-1}) = 2^m + \delta$.*

Proof. Since $y_0 = q$ is not a terminal number, we have $r \geq 1$. By the minimality of r , the terms $y_0, \dots, y_{r-1}$ are not terminal numbers. As in the proof of Lemma 2, induction gives $y_{r-1} \geq 3$ and $k_{r-1} \geq 3$. By (7),

$$y_r = k_{r-1}(2^{j_{r-1}} k_{r-1} + \varepsilon_{r-1}),$$

where $\varepsilon_{r-1} = 1$ under Q_- and $\varepsilon_{r-1} = -1$ under Q_+ by the decomposition (2). The terminal number y_r is $2^m + 1$ under Q_- , $2^m - 1$ under Q_+ , and $2^m \pm 1$ under S , with $m \geq 1$. This proves (i), (ii) and (iii). □

Section 4 combines Lemma 3 with the solutions of (3) in Lemma 5 to prove Theorems 1, 2 and 3.

3 The exponential Diophantine equation

We solve (3) with the following result of Szalay [5], in the form stated by Scott [1, Theorems 1.1 and 1.2].

Lemma 4 (Szalay [5]) *Consider the solutions of $2^a \pm 2^b + 1 = z^2$ in positive integers a, b and z .*

(i) *The solutions of $2^a + 2^b + 1 = z^2$ with $a \geq b$ are*

$$(a, b, z) = (2u, u + 1, 2^u + 1) \text{ with } u \geq 1, \quad (5, 4, 7), \quad (9, 4, 23).$$

(ii) *The solutions of $2^a - 2^b + 1 = z^2$ with $a > b$ are*

$$(a, b, z) = (2u, u + 1, 2^u - 1) \text{ with } u \geq 2, \quad (5, 3, 5), \quad (7, 3, 11), \quad (15, 3, 181).$$

Apart from textbook results, Lemma 4 is the only external result used in our proofs.

Lemma 5 *The solutions of (3) are given by the identities (5), that is, $(j, k, \varepsilon, m, \delta)$ is $(1, 3, -1, 4, -1)$, $(2, 3, -1, 5, 1)$ or $(1, 45, 1, 12, -1)$. In particular, the following hold for (3) with $\varepsilon = \delta$.*

(i) *The equation $k(2^j k + 1) = 2^m + 1$ has no solution.*

(ii) *The only solution of $k(2^j k - 1) = 2^m - 1$ is $(j, k, m) = (1, 3, 4)$.*

Proof. The identities (5) hold. Conversely, let $(j, k, \varepsilon, m, \delta)$ satisfy (3). By (4),

$$(a, b, z) = (m + j + 2, j + 2, 2^{j+1}k + \varepsilon)$$

solves the equation of Lemma 4 (i) if $\delta = 1$, and that of Lemma 4 (ii) if $\delta = -1$. Since $m \geq 1$, the exponent a exceeds b . Since $j \geq 1$, we have $z \equiv \varepsilon \pmod{4}$, so z determines ε . Then $j = b - 2$, $k = (z - \varepsilon)/2^{j+1}$ and $m = a - j - 2$.

If (a, b, z) belongs to the family $(2u, u + 1, 2^u + \delta)$ of Lemma 4, then $u = b - 1 = j + 1$, and $2^{j+1}k + \varepsilon = 2^{j+1} + \delta$ gives $2^{j+1}(k - 1) = \delta - \varepsilon$. This is impossible, since $2^{j+1}(k - 1) \geq 8$. For $\delta = 1$, the solution $(5, 4, 7)$ gives $j = 2$, $\varepsilon = -1$ and $k = 1$, which is excluded. The solution $(9, 4, 23)$ gives $(j, k, \varepsilon, m) = (2, 3, -1, 5)$. For $\delta = -1$, the solution $(5, 3, 5)$ gives $j = 1$, $\varepsilon = 1$ and $k = 1$, which is excluded. The solutions $(7, 3, 11)$ and $(15, 3, 181)$ give $(j, k, \varepsilon, m) = (1, 3, -1, 4)$ and $(1, 45, 1, 12)$. These are the identities (5). None of them has $\varepsilon = \delta = 1$, and only $(j, k, \varepsilon, m, \delta) = (1, 3, -1, 4, -1)$ has $\varepsilon = \delta = -1$. This proves (i) and (ii). $\square$

Lemma 5 (i) is the statement of [3, Lemma 2]. The next remark discusses the descent by which it is proved there.

Remark 1 *The proof of [3, Lemma 2] writes $k - 1 = 2^{\alpha_1} h_1$, then $h_1 + 1 = 2^{\alpha_2} h_2$. In general, it writes $h_i + 1$ or $h_i - 1$ as $2^{\alpha_{i+1}} h_{i+1}$ with h_{i+1} odd. Sedaghat [3] denotes α_i and h_i by j_i and k_i . At each reduction, the proof divides the equation by the least power of 2 among its terms. The reduction is impossible when $h_i - 1 = 0$, and the proof does not treat this case.*

For fixed $j, m \geq 1$, reducing $k(2^j k + 1) = 2^m + 1$ modulo any given power of 2 does not exclude a solution. For fixed $j \geq 1$ and integers k, k' ,

$$(2^j k^2 + k) - (2^j k'^2 + k') = (k - k')(2^j(k + k') + 1),$$

and the factor $2^j(k + k') + 1$ is odd. Hence the map $k \mapsto 2^j k^2 + k$ is injective, and therefore bijective, modulo every power of 2. In particular, the congruence $2^j k^2 + k \equiv 2^m + 1$ has a solution modulo every power of 2, and this solution is odd, since $2^j k^2 + k \equiv k \pmod{2}$.

We apply the same descent to $k(2^j k - 1) = 2^m - 1$, which is (3) with $\varepsilon = \delta = -1$. Write it as $2^j k^2 - k + 1 = 2^m$. Since $k \geq 3$, this gives $2^m > 2^j$. Write $k - 1 = 2^{\alpha_1} h_1$ with h_1 odd, so that

$$2^j k^2 - 2^{\alpha_1} h_1 = 2^m.$$

As k and h_1 are odd and $m > j$, comparing the exponents of 2 in the terms of this equation forces $\alpha_1 = j$. Division by 2^j gives $k^2 - h_1 = 2^{m-j}$. Substituting $k = 2^j h_1 + 1$ gives

$$2^{2j} h_1^2 + 2^{j+1} h_1 + 1 - h_1 = 2^{m-j},$$

and the next reduction would write $h_1 - 1$ as $2^{\alpha_2} h_2$. At the solution $(j, k, m) = (1, 3, 4)$ of Lemma 5 (ii), $h_1 = 1$, and the descent stops. If the descent continued indefinitely, it would exclude this solution, and hence the bounded orbits of 5 under Q_+ and of 11 under S .

4 Proofs of the classification theorems

We now prove Theorems 1, 2 and 3. Each proof first treats the initial values with a bounded orbit, and then shows that every other orbit tends to infinity. To show this, we assume that some odd term of the orbit is a terminal number. Lemma 3 then gives a solution of (3), and Lemma 5 lists all solutions. Under Q_- , no solution exists. Under Q_+ and S , the solutions force the odd part of x_0 to be 5 and to lie in $\{11, 23, 181\}$, respectively.

Proof of Theorem 1. Since $Q_-(0) = Q_-(1) = 0$, the orbits of 0 and of the powers of 2 reach 0, which gives (i). For $x_0 = 2^l(2^m + 1)$, the odd part $2^m + 1$ is a terminal number, and Lemma 2 (ii) gives (ii). Every other x_0 has an odd part $q > 1$ that is not a terminal number. If some odd term were a terminal number, Lemma 3 (i) would give a solution of (3) with $\varepsilon = \delta = 1$, which contradicts Lemma 5 (i). Hence $x_i \rightarrow \infty$ by Lemma 2 (ii). This proves (iii). $\square$

Proof of Theorem 2. Part (i) holds since $Q_+(0) = 0$. For $x_0 = 2^l(2^m - 1)$ with $m \geq 2$, the odd part $2^m - 1$ is a terminal number, and Lemma 2 (ii) applies. For $m = 1$, the orbit of $x_0 = 2^l$ reaches the fixed point 1. This gives (ii). For $x_0 = 5 \cdot 2^l$, the odd part $y = 5 = 2 \cdot 3 - 1$ has $j = 1$ and $k = 3$ in the decomposition (2), and Lemma 1 gives the terminal number $y' = 15 = 2^4 - 1$. Lemma 2 (ii) gives (iii).

Every other x_0 has an odd part $q \geq 3$ that is neither a terminal number nor 5. Assume that some odd term is a terminal number, and let r and $y_r = 2^m - 1$ be as in Lemma 3 (ii). By Lemma 5 (ii), $(j_{r-1}, k_{r-1}, m) = (1, 3, 4)$, so $y_{r-1} = 2 \cdot 3 - 1 = 5$ by the decomposition (2). By Lemma 2 (i), q divides $y_{r-1} = 5$. Since $q \geq 3$, this gives $q = 5$, which is excluded. Hence no odd term is a terminal number, and $x_i \rightarrow \infty$ by Lemma 2 (ii). This proves (iv). $\square$

Proof of Theorem 3. The orbit of 0 stays at 0. If the odd part q of x_0 is a terminal number, the orbit reaches 0 by the discussion before Lemma 2. If $y = q$ is $11 = 4 \cdot 3 - 1$, $23 = 8 \cdot 3 - 1$ or $181 = 4 \cdot 45 + 1$, written as in the decomposition (2), Lemma 1 gives $y' = 15 = 2^4 - 1$, $33 = 2^5 + 1$ or $4095 = 2^{12} - 1$, respectively. Then y' is a terminal number, and the orbit reaches 0 by Lemma 2 (ii). This proves (i).

For (ii), let $x_0 \geq 1$ have an odd part q that is neither a terminal number nor in $\{11, 23, 181\}$. Then $q > 1$, since $1 = 2^1 - 1$ is a terminal number. Assume that some odd term is a terminal number, and let r be as in Lemma 3 (iii). By Lemma 5 and the decomposition (2), the term $y_{r-1} = 2^{j_{r-1}+1}k_{r-1} + \varepsilon_{r-1}$ is 11, 23 or 181, and each of these numbers is prime. If $r \geq 2$, then (7) gives $y_{r-1} = k_{r-2}(2^{j_{r-2}}k_{r-2} + \varepsilon_{r-2})$. As in the proof of Lemma 3, $k_{r-2} \geq 3$, so $2^{j_{r-2}}k_{r-2} + \varepsilon_{r-2} \geq 5$ and y_{r-1} is composite. Hence $r = 1$ and $q = y_0$ lies in $\{11, 23, 181\}$, a contradiction. So no odd term is a terminal number, and Lemma 2 (ii) proves (ii). $\square$

Theorems 1, 2 and 3 have the following consequences.

Corollary 1 (i) For $X \geq 2$, at most $(\log_2 X + 1)^2$ initial values in $\{0, 1, \dots, X\}$ have a bounded orbit under Q_- , and the same bound holds under Q_+ .

(ii) For $X \geq 2$, at most $(\log_2 X + 3)^2$ initial values in $\{0, 1, \dots, X\}$ have a bounded orbit under S .

(iii) An orbit under Q_- that contains an odd term $2^m - 1$ with $m \geq 3$ tends to infinity.

(iv) An orbit under Q_+ that contains an odd term $2^m + 1$ with $m \geq 3$ tends to infinity.

Proof. Under Q_- , by Theorem 1, a bounded orbit starts from 0, from a power $2^l \leq X$, or from $2^l(2^m + 1) \leq X$. There are at most $\log_2 X + 1$ powers $2^l \leq X$. For the initial values $2^l(2^m + 1) \leq X$,

$$2^{l+m} < 2^l(2^m + 1) \leq X, \quad \text{so} \quad l + m < \log_2 X,$$

and at most $\log_2 X (\log_2 X + 1)/2$ pairs (l, m) with $l \geq 0$ and $m \geq 1$ qualify. Since $\log_2 X \geq 1$, the total is at most

$$1 + (\log_2 X + 1) + \frac{\log_2 X (\log_2 X + 1)}{2} \leq (\log_2 X + 1)^2.$$

Under Q_+ , by Theorem 2, a bounded orbit starts from 0, from $2^l(2^m - 1) \leq X$, or from $5 \cdot 2^l \leq X$. For the initial values $2^l(2^m - 1) \leq X$,

$$2^{l+m-1} \leq 2^l(2^m - 1) \leq X, \quad \text{so} \quad l + m \leq \log_2 X + 1,$$

and at most $(\log_2 X + 1)(\log_2 X + 2)/2$ pairs (l, m) qualify. For $5 \cdot 2^l \leq X$, we have $2^{l+2} < X$, which leaves at most $\log_2 X - 1$ values of l . Since $\log_2 X \geq 1$, the total is at most

$$1 + \frac{(\log_2 X + 1)(\log_2 X + 2)}{2} + \log_2 X - 1 \leq (\log_2 X + 1)^2.$$

This proves (i).

Under S , by Theorem 3, a bounded orbit starts from 0, from $2^l(2^m - 1) \leq X$, from $2^l(2^m + 1) \leq X$, or from $2^l c \leq X$ with $c \in \{11, 23, 181\}$. The counts above bound the families $2^l(2^m - 1)$ and $2^l(2^m + 1)$ by $(\log_2 X + 1)(\log_2 X + 2)/2$ and $\log_2 X (\log_2 X + 1)/2$, respectively. For each c , we have $2^{l+3} < 2^l c \leq X$, which leaves at most $\log_2 X$ values of l . The total is at most

$$1 + \frac{(\log_2 X + 1)(\log_2 X + 2)}{2} + \frac{\log_2 X (\log_2 X + 1)}{2} + 3 \log_2 X \leq (\log_2 X + 3)^2.$$

This proves (ii).

Under Q_- , from the odd term $2^m - 1$ on, the orbit is the orbit of the initial value $2^m - 1$. For $m \geq 3$, this number is at least 7 and congruent to 3 modulo 4, whereas every terminal number of Q_- that is at least 7 is congruent to 1 modulo 4. So $2^m - 1$ is neither a terminal number nor a power of 2, and Theorem 1 (iii) applies. This proves (iii).

Under Q_+ , from the odd term $2^m + 1$ on, the orbit is the orbit of the initial value $2^m + 1$. For $m \geq 3$, this number is at least 9 and congruent to 1 modulo 4, whereas every terminal number of Q_+ that is at least 9 is congruent to 3 modulo 4. So $2^m + 1$ is neither a terminal number nor 5, and Theorem 2 (iv) applies. This proves (iv). $\square$

Corollary 1 (iii) proves [2, Conjecture 4]. In Corollary 1 (iii) and (iv), the condition $m \geq 3$ cannot be relaxed. Under Q_- , the orbit of $2^1 - 1 = 1$ reaches 0, and $2^2 - 1 = 3$ is a fixed point. Under Q_+ , the number $2^1 + 1 = 3 = 2^2 - 1$ lies on the cycle $3 \rightarrow 6 \rightarrow 3$, and the orbit of $2^2 + 1 = 5$ is bounded by Theorem 2 (iii). Under S , every number $2^m \pm 1$ is a terminal number, so Corollary 1 (iii) and (iv) have no counterpart for S .

5 A divisibility criterion for Q_-

The proof of Theorem 1 relies on Lemma 4. In this section, we give a criterion for unbounded orbits under Q_- that uses only divisibility. For an odd integer $n > 1$, let $\text{ord}_n(2)$ be the multiplicative order of 2 modulo n , the least $d \geq 1$ with $2^d \equiv 1 \pmod{n}$. For an integer $d \geq 1$, let $v_2(d)$ be its 2-adic valuation, the exponent of the largest power of 2 dividing d .

Theorem 4 (i) *Let $x_0 \geq 1$, and let $y > 1$ be an odd term of its orbit $(x_i)_{i \geq 0}$ under Q_- . If y divides none of the numbers $2^m + 1$ with $m \geq 1$, then $x_i \rightarrow \infty$ as $i \rightarrow \infty$.*

(ii) *An odd integer $y > 1$ divides $2^m + 1$ for some $m \geq 1$ if and only if there is $s \geq 1$ with $v_2(\text{ord}_p(2)) = s$ for every prime p dividing y .*

Proof. Let i_0 be an index with $x_{i_0} = y$. Lemma 2 (i), applied to the orbit of y , shows that y divides x_i for every $i \geq i_0$. If x_i does not tend to infinity, Lemma 2 (ii), applied to the orbit of y , gives an index $i_1 \geq i_0$ and an integer $m \geq 1$ with $x_{i_1} = 2^m + 1$. Then y divides $2^m + 1$, which contradicts the assumption on y . This proves (i).

For (ii), let p be a prime dividing y , and let p^e be the largest power of p dividing y . We show that, for every $m \geq 1$,

$$2^m \equiv -1 \pmod{p^e} \iff \text{ord}_{p^e}(2) \text{ divides } 2m \text{ and not } m.$$

If $2^m \equiv -1 \pmod{p^e}$, then $2^{2m} \equiv 1$ and $2^m \not\equiv 1 \pmod{p^e}$. This gives the forward implication. Conversely, assume that $\text{ord}_{p^e}(2)$ divides $2m$ and not m . Then p^e divides the product $(2^m - 1)(2^m + 1)$ but not $2^m - 1$. These factors differ by 2, so p does not divide both, and p^e divides $2^m + 1$. Equivalently, the condition on $\text{ord}_{p^e}(2)$ reads

$$v_2(m) = v_2(\text{ord}_{p^e}(2)) - 1 \quad \text{and} \quad \text{the odd part of } \text{ord}_{p^e}(2) \text{ divides } m. \quad (8)$$

Moreover, $v_2(\text{ord}_{p^e}(2)) = v_2(\text{ord}_p(2))$. Indeed, $\text{ord}_p(2)$ divides $\text{ord}_{p^e}(2)$, since $2^{\text{ord}_{p^e}(2)} \equiv 1 \pmod{p}$. By the binomial theorem, if an integer is congruent to 1 modulo p^f with $f \geq 1$, its p -th power is congruent to 1 modulo p^{f+1} . Applying this $e - 1$ times, starting from $2^{\text{ord}_p(2)} \equiv 1 \pmod{p}$, gives

$$2^{\text{ord}_p(2)p^{e-1}} \equiv 1 \pmod{p^e},$$

so $\text{ord}_{p^e}(2)$ divides $\text{ord}_p(2)p^{e-1}$. The quotient $\text{ord}_{p^e}(2)/\text{ord}_p(2)$ therefore divides the odd number p^{e-1} , and the 2-adic valuations agree.

Now y divides $2^m + 1$ if and only if (8) holds for every prime p dividing y . If such an m exists, then $v_2(\text{ord}_p(2)) = v_2(m) + 1$ for every prime p dividing y . Conversely, assume that $v_2(\text{ord}_p(2)) = s \geq 1$ for every such p . Let m be 2^{s-1} times the least common multiple of the odd parts of the orders $\text{ord}_{p^e}(2)$. Then m satisfies (8) for every p . This proves (ii). $\square$

We now apply Theorem 4 to the conjectures of Sedaghat [2]. For $m \geq 3$, the residues of the powers of 2 modulo $2^m - 1$ are $2^0, 2^1, \dots, 2^{m-1}$, and none of them is congruent to -1 , since $2^{m-1} < 2^m - 2$. Hence $2^m - 1$ divides no number $2^{m'} + 1$, and Theorem 4 (i) applies to every odd term $2^m - 1$. This proves Corollary 1 (iii) without Lemma 4. With $x_0 = 7 = 2^3 - 1$, Theorem 4 (i) also proves that unbounded orbits exist, which is [2, Conjecture 3]. In [3], both conjectures are derived from the classification stated there.

Without Lemma 4, Theorem 4 also shows that the initial values with a bounded orbit under Q_- have natural density 0, that is, only $o(X)$ of them lie in $\{0, 1, \dots, X\}$. This was announced in [3]. Let $p \equiv 7 \pmod{8}$ be prime. By Euler's criterion and the supplementary law of quadratic reciprocity for 2 [4, Theorem 12.1 (i) and (iv)], $2^{(p-1)/2} \equiv 1 \pmod{p}$. Since $(p-1)/2$ is odd, $v_2(\text{ord}_p(2)) = 0$, and by Theorem 4 (ii) no odd multiple of p divides a number $2^m + 1$. By Theorem 4 (i), applied to the odd part of x_0 , every $x_0 \geq 1$ with a prime factor $p \equiv 7 \pmod{8}$ has an unbounded orbit.

For a finite set P of primes $p \equiv 7 \pmod{8}$, inclusion-exclusion shows that the number of $x_0 \in \{1, \dots, X\}$ divisible by no $p \in P$ is

$$X \prod_{p \in P} \left(1 - \frac{1}{p}\right) + O(2^{|P|}). \quad (9)$$

By the prime number theorem for arithmetic progressions [4, Theorem 5.21], the number of primes $p \leq X$ with $p \equiv 7 \pmod{8}$ is asymptotic to $X/(4 \log X)$ as $X \rightarrow \infty$, and partial summation shows that the sum of $1/p$ over these primes diverges. Hence the product $\prod_{p \in P} (1 - 1/p)$ in (9) can be made arbitrarily small by enlarging P . Every $x_0 \geq 1$ with a bounded orbit is divisible by no $p \in P$. So, for every such P , the number of these initial values in $\{1, \dots, X\}$ is at most (9), and therefore this number is $o(X)$. Corollary 1 (i) gives the much stronger bound $(\log_2 X + 1)^2$, but its proof uses Lemma 4.

The criterion does not carry over to Q_+ and S . Under Q_+ , Lemma 2 shows that a bounded orbit of $x_0 = 2^l q$ with $q > 1$ enters a cycle that contains a number $2^m - 1$ divisible by q . The analogue of Theorem 4 (i) for Q_+ would state that an orbit with an odd term $y > 1$ that divides none of the numbers $2^m - 1$ tends to infinity. But every odd $y > 1$ divides $2^{\text{ord}_y(2)} - 1$, so this statement applies to no orbit. Under S , the numbers $(y-1)/2$ and $(y+1)/2$ are coprime to an odd $y > 1$, so y does not divide $S(y)$, and Lemma 2 (i) has no analogue.

The proof of Theorem 2 uses Lemma 5 only through (ii), which concerns $\delta = -1$ and hence needs only Lemma 4 (ii). According to Scott [1, Section 1], Szalay's proof of Lemma 4 (ii) uses a non-elementary bound of Beukers, and replacing this bound by an earlier elementary theorem of Beukers makes the proof elementary. Hence Theorem 2 has an elementary proof. The proof of Theorem 3 uses both parts of Lemma 4.

6 Conclusion

We leave the following questions open.

Under Q_- , does some initial value not covered by Theorem 1 (i) and (ii) have an orbit in which every odd term divides some number $2^m + 1$? By Theorem 4 (i), an elementary proof that no such initial value exists would give an elementary proof of Theorem 1 (iii). Is there an elementary proof that (3) has no solution with $\varepsilon = \delta = 1$? Such a proof would make Theorem 1 elementary.

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